# Live Demo Flow & Presenter Guide

Step-by-Step Walkthrough, System Explanations & Talking Points

12–15 Min Live Walkthrough

## Pre-Demo Setup (30 Seconds Before Sharing Screen)

`http://127.0.0.1:5173`

**1. Main Assurance UI**

Start via `./start.sh`. Click **Reset** in the top bar so status reads **Healthy**. Optional: click slider icon to set a Claude/GPT key.

`http://127.0.0.1:8001`

**2. Multi-OEM Lab Emulator**

Keep open in Browser Tab 2 for technical calls to show real ServiceNow, Jira, WFS GeoJSON & binary `.sor` feeds.

`http://127.0.0.1:8002/docs`

**3. FastAPI OpenAPI Docs**

Optional Tab 3 for solution architects to inspect typed REST endpoints, PostGIS queries & correlation schemas.

## Tailoring the Flow by Operator Profile (Top-Right Selector)

| OPERATOR ARCHETYPE | DROPDOWN PROFILE TO SELECT | LEAD SCENARIOS TO EMPHASISE |
|---|---|---|
| **Metro Fibre & DCI** (e.g., Constl) | **Pure-Play Metro Fibre, Wavelength & DCI** | Act 1 (Shared-duct audit) → Act 2 (Canal trunk cut + 100G reroute) → DIFC Bank shared chamber → Pre-failure BER. |
| **National Converged** (e.g., e&, du, Vi) | **Converged 4G/5G + Fibre + Enterprise** | Act 2 (13-alarm storm across optical/IP/5G) → Act 4 (AI Quoz power blackout, 5G battery drain & 80 GHz rain fade). |
| **Subsea & Wholesale** (e.g., PCCW Global) | **Global Subsea, Optical DCI & Wholesale** | Act 1 (Diverse DCI route proof) → Act 4C (Jebel Ali CLS subsea repeater shunt) → Act 2 (100G protection switch). |

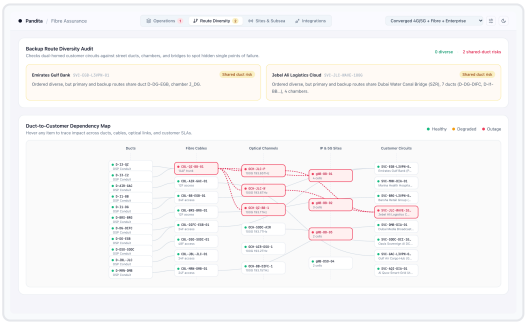
## ACT 1 Proactive Route Diversity & Blast-Radius Audit

2 Minutes

**1. Click to Open:** On a healthy network, point to **2 shared-duct risks →** on the top status bar (or click **Route Diversity** in the top navigation).

**2. What Is Happening Under the Hood:** Even before an alarm fires, the graph engine cross-checks BSS customer circuits marked *"Ordered Diverse"* against PostGIS underground street ducts, bridges, and manhole chambers. It catches two Platinum circuits (**Emirates Gulf Bank** and **Jebel Ali Logistics Cloud**) whose primary and backup legs quietly cross the same chamber (`J_DG`) or bridge (**Dubai Water Canal Bridge**).

**3. Interactive Dependency Trace:** Hover over any box in the **Duct-to-Customer Dependency Map** below to highlight the 5-layer chain: *OSP Conduit → Fibre Trunk → 100G Optical Channel → 5G Tower → Enterprise SLA*.



Route Diversity tab: Shared-duct audit + 5-layer blast-radius map

> **What to Say:** "Enterprise and hyperscale customers pay for dual-homed diversity, yet logical circuit inventories rarely check street-level civil ducts. Before any outage happens, we expose hidden single points of failure—and let pre-sales teams verify truly diverse DCI/RFP paths in seconds instead of days."

## ACT 2 Street-Accurate Fibre-Cut Pinpointing & Gated Recovery

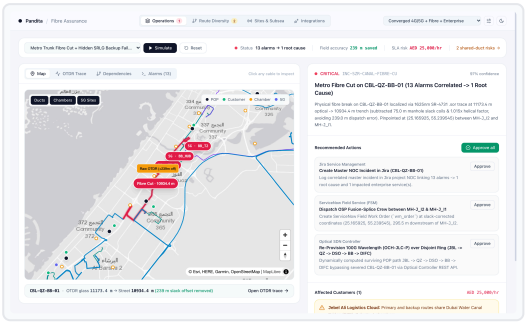
3.5 Minutes

**1. Trigger Incident:** Click **Operations**, select **Metro Trunk Fibre Cut + Hidden SRLG Backup Failure** in the top bar, and click **▶ Simulate**.

**2. Multi-Vendor Storm Collapse:** Point to the status bar: **13 alarms → 1 root cause** (97% confidence). Optical LOS (Ciena), router BFD drops (Cisco/Juniper), VIAVI OTDR, and 3 cascading 5G sites (`BB_HUB`, `BB_T1`, `BB_T2`) are grouped onto trunk cable `CBL-QZ-BB-01`, surfacing **AED 25,000/hr** SLA risk immediately.

**3. Raw Glass vs. Street Trench GPS (239 m Saved):** On the map, contrast the amber pin (**Raw OTDR (+239m off)**) against the red pin (**Fibre Cut · 10934.4 m**). *Explain the physics:* OTDR measures time-of-flight through glass (`11,173.4 m`), which includes `75 m` of coiled repair slack inside upstream manholes plus `1.015x` cable twist. Subtracting both pinpoints the exact dig spot between `MH-J_I2` and `MH-J_I1`.

**4. Human-Approved Execution:** Review the 3 staged actions on the right (**Jira** master ticket, **ServiceNow FSM** splice dispatch with GPS, and **Optical SDN** 100G disjoint reroute). Click **Approve all** to execute live.



Operations Map: Raw vs. slack-adjusted GPS + 1-click NOC approvals

> **What to Say:** "Without manhole-slack correction, field crews dig 200 to 300 metres away from the break—wasting 3 hours on highway permits and trenching. Here, NOC collapses 13 alarms into one street coordinate and restores the 100G circuit in one engineer-approved click."

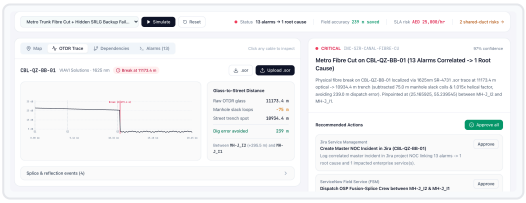
## ACT 2B Inspecting the Binary Telcordia .sor Trace & Runbook RAG

1.5 Minutes

**1. Open Waveform:** Click **Open OTDR trace →** below the map (or click **OTDR Trace**).

**2. Explain Binary Parsing:** The backend parses raw binary `Telcordia SR-4731 Rev 2 (.sor)` bytes directly from VIAVI/EXFO test heads—plotting the `1625 nm` step-drop at `11,173.4 m` alongside the **Glass-to-Street Distance** table and expandable splice-loss events.

**3. Vector RAG SOPs:** Scroll the right panel to expand **Runbook & engineering notes**—showing the `pgvector`-retrieved carrier SOP (`RB-OSP-001`) and historical resolved tickets.



OTDR Trace view: 1625 nm waveform + Glass-to-Street math

## ACT 3 Proving Zero Hardcoding: Ad-Hoc Cut & .sor Round-Trip

1.5 Minutes

**1. Live Interactive Cable Cut:** Click **Reset** → select the **Map** tab → **click any fibre cable directly on the street map**. Drag the cut-position slider (15%–88%) to any random spot and click **Simulate Cut**. The emulator synthesises a brand-new binary `.sor` waveform at that exact metre mark, PostGIS walks upstream manholes dynamically, and NetworkX computes a fresh surviving optical path.

**2. Live File Upload Proof:** On **OTDR Trace**, click **↓ .sor** to download the binary trace to your laptop, then click **Upload .sor** to re-parse the raw bytes live in front of the audience.

## ACT 4 Stopping False Fibre Truck Rolls: Power, Weather & Subsea

4 Minutes

Up to 35% of emergency OSP contractor callouts arrive on site only to discover the fibre is intact—the real culprit was a customer power trip, tower battery depletion, or heavy rain on microwave links. Walk through these four scenarios to showcase cross-domain disambiguation:

### 4A. Building Power Blackout vs. Fibre Cut (Al Quoz GPON / Enterprise)

Saves Wasted Splice Van

**Click:** Select **Premises Power Blackout ('Dying Gasp') vs. Fibre Cut Disambiguation** → **▶ Simulate**.
**Explanation:** 24 ONTs/modems drop within 1 second. Instead of blindly rolling a fibre crew, the classifier correlates `ITU-T G.984` *dying-gasp* power-fail frames (≥80% cohort match) with an automated clean `1625 nm` OTDR continuity check (*Fibre intact*). Top bar displays **Field accuracy: Trip saved**, suppressing OSP dispatch and logging a commercial power advisory instead.

### 4B. Proactive Pre-Failure Optical BER Degradation (Silicon Oasis AI DC)

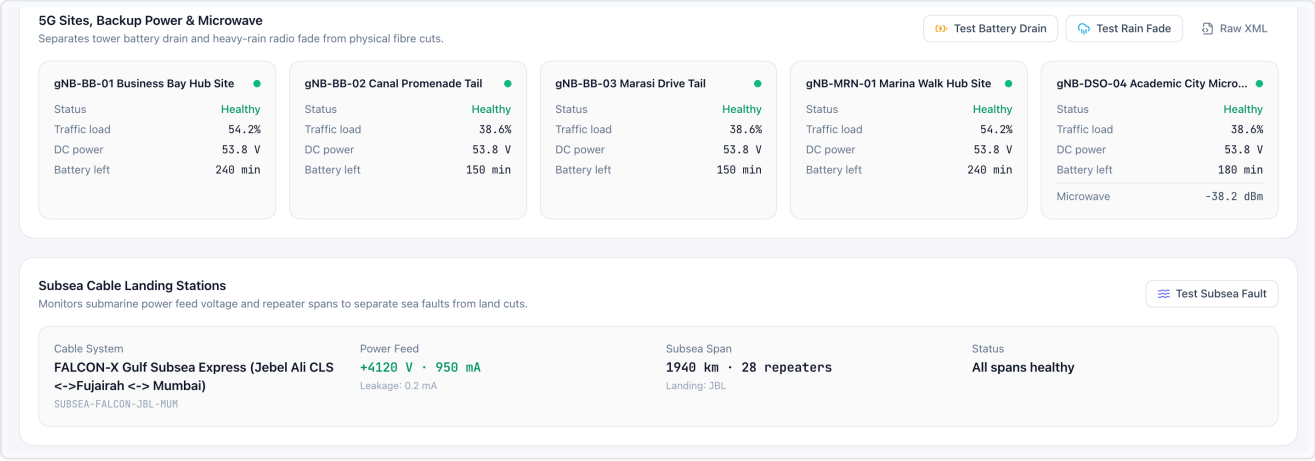
0 ms Outage / AED 0 Penalty

**Click:** Select **Pre-Failure DWDM Optical OSNR & Pre-FEC BER Degradation** → **▶ Simulate**.
**Explanation:** Spots Pre-FEC BER crossing `1e-4` alongside a non-reflective `1.8 dB` splice pinch on the OTDR trace *while Post-FEC errors are still zero*. Stages a proactive hitless wavelength shift and preventive chamber maintenance before the customer sees a single dropped packet.

### 4C. 5G Hub Battery Countdown, 80 GHz Rain Fade & Subsea Shunt

Sites & Subsea Tab

**Click:** Open **Sites & Subsea** in the top nav and trigger each domain test button:
• **Test Battery Drain** — Rectifier AC mains fail on `gNB-BB-01`; tracks DC voltage slope toward `43.2 V` cutoff (`36 min` reserve) and dispatches a mobile diesel generator before 9 daisy-chained 5G cells go dark (confirming fibre backhaul is intact).
• **Test Rain Fade** — Correlates `-38.2 dBm` RSL fade & `1024QAM → QPSK` modulation downshift on `80 GHz` E-band hop with convective rain, holding ticket in auto-clear watch instead of dispatching tower climbers.
• **Test Subsea Fault** — Detects `+4120 V` PFE voltage drop & C-OTDR insulation shunt on **FALCON-X Gulf Subsea Express** (`14.2 km` seaward of Jebel Ali CLS), routing wavelengths onto protection paths without rolling land OSP crews.




Sites & Subsea tab: Live 5G DC power, battery reserve, E-band microwave RSL & FALCON-X submarine PFE telemetry

## ACT 5 Multi-OEM Connectors, 10k-Alarm Stress Test & API Proof

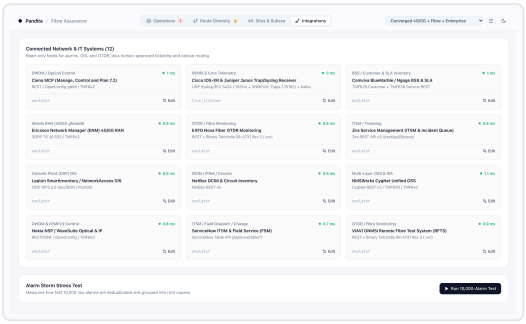
2 Minutes

**1. 12 Live Connectors:** Click `Integrations` in the top nav. Walk across the 12 standards-based adapters: **Ciena MCP**, **Nokia NSP**, **VIAVI ONMSi**, **EXFO Nova**, **Lepton GIS**, **Ericsson ENM**, **Cisco/Juniper Syslog & SNMP**, **NetBox**, **Comviva BSS**, **NMSWorks**, **ServiceNow FSM**, and **Jira SM**.

**2. Zero-Code Cutover:** Click `Edit` on any card to show how each connector toggles cleanly from `emulator` to a `live` carrier endpoint URL.

**3. 10,000-Alarm Stress Test:** Click `▶ Run 10,000-Alarm Test` at the bottom to benchmark real-time deduplication, flap suppression, and causal graph clustering in milliseconds.

**4. Optional Wire Proof (:8001):** Switch to Browser Tab 2 (`http://127.0.0.1:8001`) and click `/servicenow/api/now/table/wm_order` or `/jira/rest/api/3/issue` to show the real JSON tickets created when you clicked *Approve all* in Act 2.



Integrations tab: 12 vendor adapters + 10,000-alarm storm test

### Bonus Financial-Sector Scenario: Dual-Homed Bank HQ Outage via Shared Manhole (DIFC Gate)

Banking & DCI Calls

**Click:** In `Operations`, select `Dual-Homed Bank HQ Outage via Shared Manhole Chamber (DIFC Gate SRLG Violation)` → `▶ Simulate`.

**Explanation:** Demonstrates what happens when civil excavation breaches chamber `J_DG`—taking down both the primary and "diverse" backup L3VPN fibres for **Emirates Gulf Bank** simultaneously—proving visually why Act 1's proactive shared-duct audit is critical before signing Platinum SLAs.

### Presenter Q&A Cheat Sheet (Most Common Buyer & CTO Questions)

**1. "Does AI guess the fault GPS or optical backup route?"**

Never. Dig coordinates, manhole slack subtraction, and Zero-SRLG backup paths are computed deterministically by binary `SR-4731 (.sor)` math, `PostGIS`, and `NetworkX`. AI handles noisy signal classification, multi-step tool investigation, `pgvector` SOP retrieval, RFP parsing, and cited EN/AR reporting.

**2. "Can we run this air-gapped inside our own data centre?"**

Yes. The entire stack runs in `Docker` or `Kubernetes (OpenShift)` on-prem with zero subscriber PII. It supports cloud APIs (`Claude Sonnet 4 / GPT-4o`), self-hosted open-weight models (`Llama 3.3 / Qwen 2.5` via `vLLM`), and a deterministic CPU-only fallback.

**3. "Will it push changes to live optical or router gear on its own?"**

No. Day-1 deployment is strictly read-only across alarms, OTDR, and GIS. Northbound actions (Ciena/Nokia wavelength switches, ServiceNow field dispatches, customer RFOs) are staged as drafts and execute only after explicit NOC engineer sign-off, logged in an immutable audit trail.

**4. "What if our GIS manhole slack records aren't clean everywhere?"**

We don't boil the ocean. We start with a **4-week single-corridor pilot** on one high-value metro ring or DCI route, ingesting your existing WFS/GeoJSON ducts and calibrating chamber slack coils automatically whenever field splices are verified.